

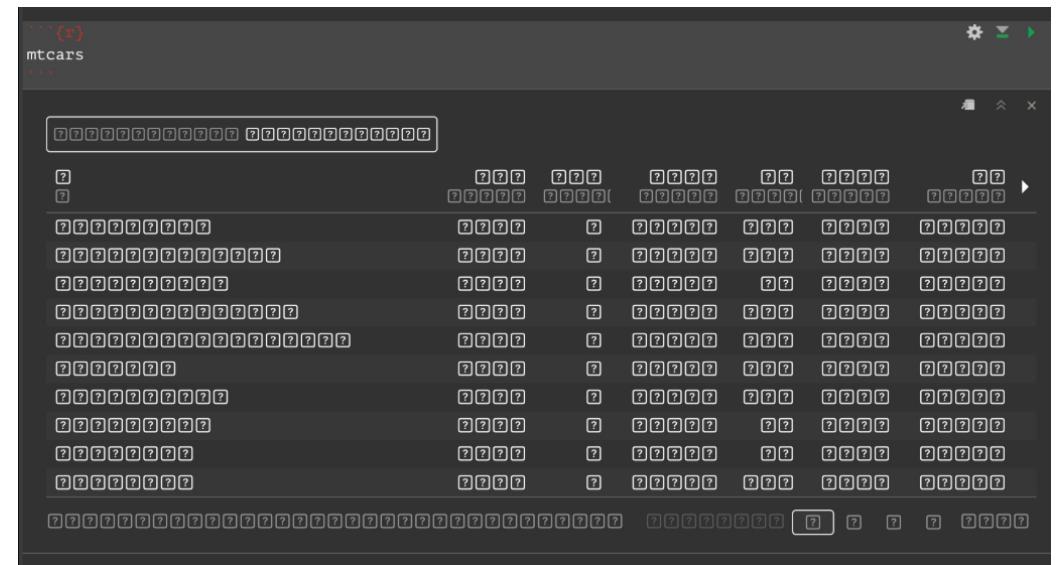
Problem Set 04 Discussion

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General Notes

- Problem set 02-04 grades + comments uploaded to Canvas
 - Future assignments added with due dates
- MacOS 26 Tahoe update – Quarto text rendering error
 - Redownload new version of the Lucida Sans font to fontbook



Formatting

- Reminder to include the **html: embed-resources = TRUE** header
 - The rendered HTML output will look correct on your computer but the plots + data won't upload to github!
 - Use the header from the problem set template notebook

Default header when creating
new Quarto document

```
---  
title: "Problemset_03"  
format: html  
editor: visual  
---
```

Header with embed-resources
from Problem Set Template

```
---  
title: "Problem Set XX"  
author: "Your Name, NetID"  
date: today  
date-format: iso  
format:  
  html:  
    embed-resources: true  
editor: source  
---
```

Slicing – multiple options

```
penguins |>  
  slice(seq(1, n(), by = 5))
```

```
penguins |>  
  slice(seq(1, nrow(penguins), by = 5))
```

```
penguins |>  
  slice(seq(from=1, to=344, by=5))
```

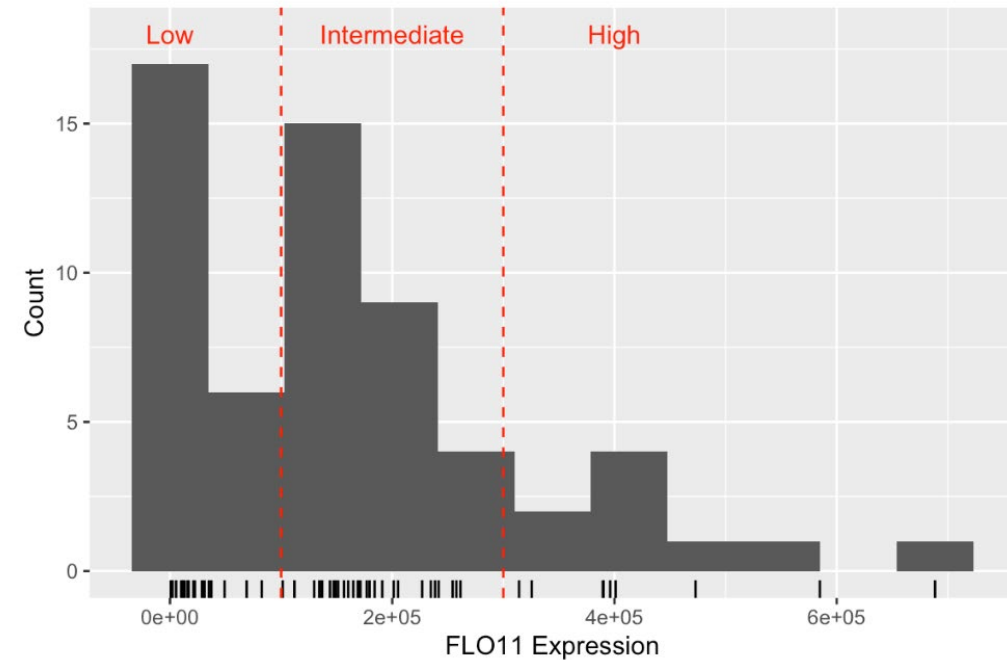
- Multiple ways to specify all the rows!
 - `n()` = returns the number of rows in the **pipd** data frame/group
 - `dplyr` function – used within `mutate()`, `slice()`, etc.
 - group-aware!
 - `nrow(penguins)` = returns the number of rows in **any** data frame (specified here as `penguins`)
 - Base R function – can be used anywhere
 - Always acts on the whole data frame, not groups
 - 344 = known number of rows
 - Hardcoded – will not change if the `penguins` table changes

Breaks – multiple options

```
segdata |>
  mutate(Flo11.group = cut(Flo11.expr,
    breaks = c(0, 1e+05, 3e+05, 7e+05),
    labels = c("Low", "Intermediate", "High"),
    ordered_result = TRUE)
```

```
yeast |>
  mutate(Flo11.group = cut(Flo11.expr,
    breaks = c(0, 100000, 300000, 800000),
    labels = c("low", "intermediate", "high"),
    ordered_results = TRUE)
```

```
yeast |>
  mutate(Flo11.group = cut(Flo11.expr,
    breaks = c(0, 1e5, 3e5, Inf),
    labels = c("low", "Intermediate", "high"),
    ordered_result = TRUE)
```



cut() in R creates intervals that are **left-open** and **right-closed**

(0, 100,000], - 0 not included, 100,000 is
(100,000, 300,000], - 100,000 not included, 300,000 is
(300,000, 800,000] – 300,000 not included, but 800,000 **is** included